

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES
DATE:	

AIM: Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- User Signup & Login
- Student Enrollment in Courses
- Assignment Submission & Evaluation
- Viewing Grades and Attendance
- Admin Role Management and Report Generation

2. Define User Interactions

- Each test case simulates a real user behavior (e.g., logging in, submitting an assignment, viewing results).

3. Design Happy Path Test Cases

- Focused on validating that all core functionalities work correctly under normal conditions.
- Example: Student registers and logs in, enrolls in a course, submits assignment, and views grades.

4. Design Error Path Test Cases

- Simulate invalid inputs, system issues, or failed actions to ensure proper error handling.
- Example: Login with wrong credentials, submission without attachment, unauthorized access to admin panel.

5. Break Down Steps and Expected Results

- Each test case includes a clear sequence of actions and expected results.
- Ensures both manual testers and automation tools can follow the process easily.

6. Use Clear Naming and IDs

- Test cases are uniquely identified (e.g., TC01 – Student Login Success, TC12 – Invalid Assignment Submission).
- Facilitates easy mapping to features and tracking in Azure DevOps.

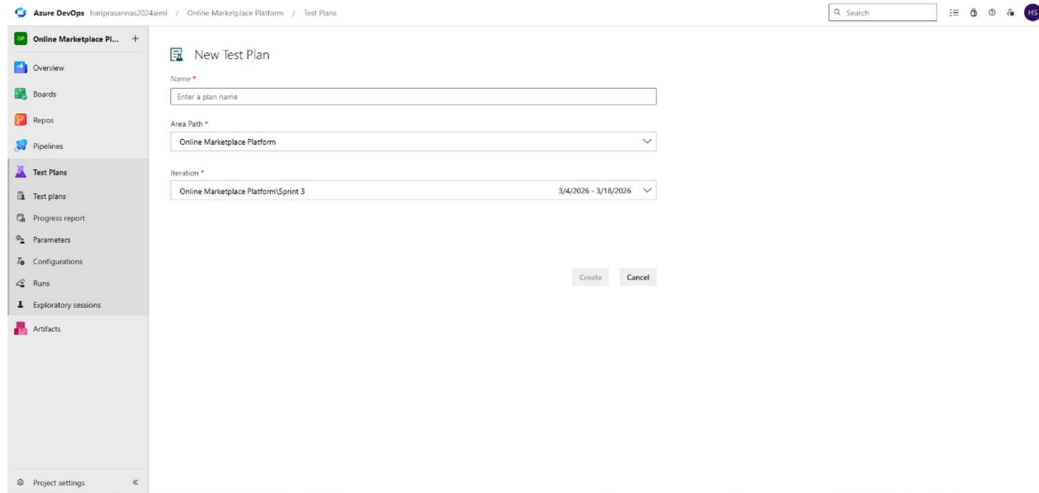
7. Separate Test Suites

- Grouped by functionality such as:
 - Login and Registration
 - Course Enrollment
 - Assignment Submission
 - Report Generation
 - Admin Functions
- Improves organization and enables focused execution in Azure DevOps.

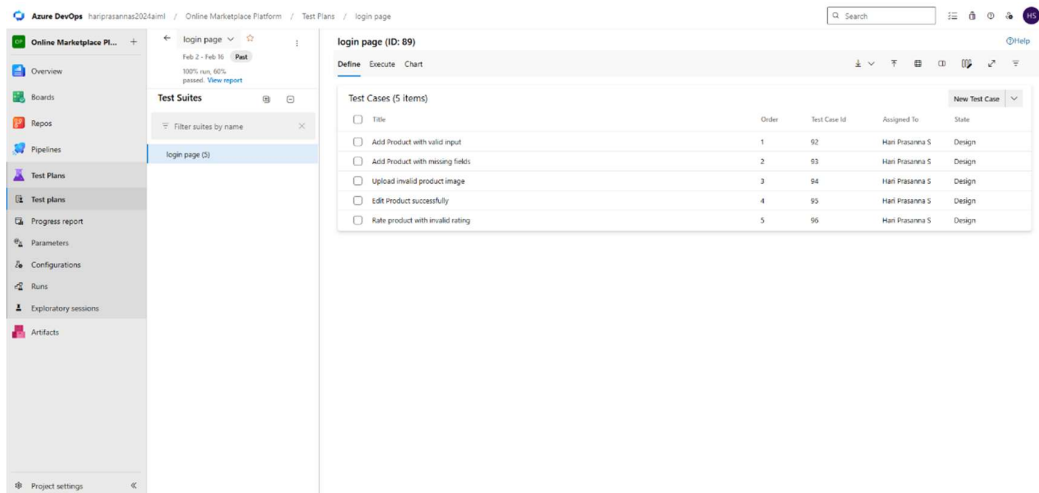
8. Prioritize and Review

- High-priority assigned to critical workflows like login, course access, and grading.
- Reviewed for completeness, accuracy, and alignment with user stories and feature definitions.

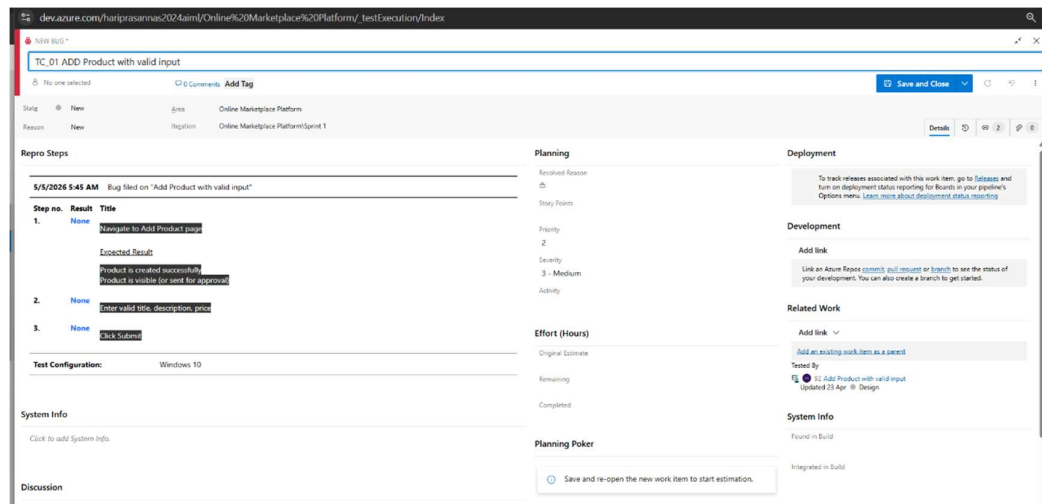
1. New test plan



2. Test suite



3. Test case:



Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Online Marketplace Platform– Test Plans

USER STORIES:

- As a student, I want to sign up and log in securely so I can access academic services. (ID: 101)
- As a student, I need to enroll in courses to participate in classes. (ID: 102)
- As a student, I should be able to submit assignments online for evaluation. (ID: 103)
- As a student, I want to view grades and attendance in one place. (ID: 104)
- As an admin, I want to manage users and generate performance reports. (ID: 105)

Test Suites:

TC_01: Student Login (Valid Credentials)

User Story ID: 101

Steps:

1. Open login page
2. Enter valid student email and password
3. Click login button
4. Verify dashboard is displayed

Expected Result: Student logs in successfully and is redirected to dashboard

Type: Happy Path

TC_02: Student Login (Invalid Credentials)

User Story ID: 101

Steps:

1. Open login page
2. Enter incorrect email or password
3. Click login button
4. Verify error message is displayed

Expected Result: Error message "Invalid username or password" is shown

Type: Error Path

TC_03: Course Enrollment (Valid)

User Story ID: 102

Steps:

1. Login to the system
2. Navigate to course list
3. Select available course
4. Click enroll

Expected Result: Enrollment confirmation is displayed

Type: Happy Path

TC_04: Course Enrollment (Course Full)

User Story ID: 102

Steps:

1. Login to the system
2. Select a course with no available seats
3. Click enroll
4. Verify system response

Expected Result: Error message "Course full" is displayed

Type: Error Path

TC_05: Assignment Submission (Valid)

User Story ID: 103

Steps:

1. Login to the system
2. Navigate to assignments
3. Upload assignment file
4. Click submit

Expected Result: Assignment submitted successfully

Type: Happy Path

TC_06: Assignment Submission (No File)

User Story ID: 103

Steps:

1. Open assignment submission page
2. Do not upload any file
3. Click submit
4. Verify error message

Expected Result: Message "Please upload a file before submitting"
Type: Error Path

TC_07: View Grades (Valid)

User Story ID: 104

Steps:

- 1. Login to the system**
- 2. Navigate to grades section**
- 3. View grades**
- 4. Verify data is displayed**

Expected Result: Grades are displayed correctly

Type: Happy Path

TC_08: View Grades (Network Failure)

User Story ID: 104

Steps:

- 1. Disconnect internet connection**
- 2. Navigate to grades section**
- 3. Attempt to load data**
- 4. Verify error message**

Expected Result: "Unable to fetch data" message is shown

Type: Error Path

TC_09: Admin Role Assignment (Valid)

User Story ID: 105

Steps:

- 1. Login as admin**
- 2. Navigate to user management**

3. Select user and assign role

4. Save changes

Expected Result: Role assigned successfully

Type: Happy Path

TC_10: Report Generation (No Data)

User Story ID: 105

Steps:

1. Login as admin

2. Navigate to reports section

3. Select empty dataset/semester

4. Click generate report

Expected Result: Message "No data available" is displayed

Type: Error Path

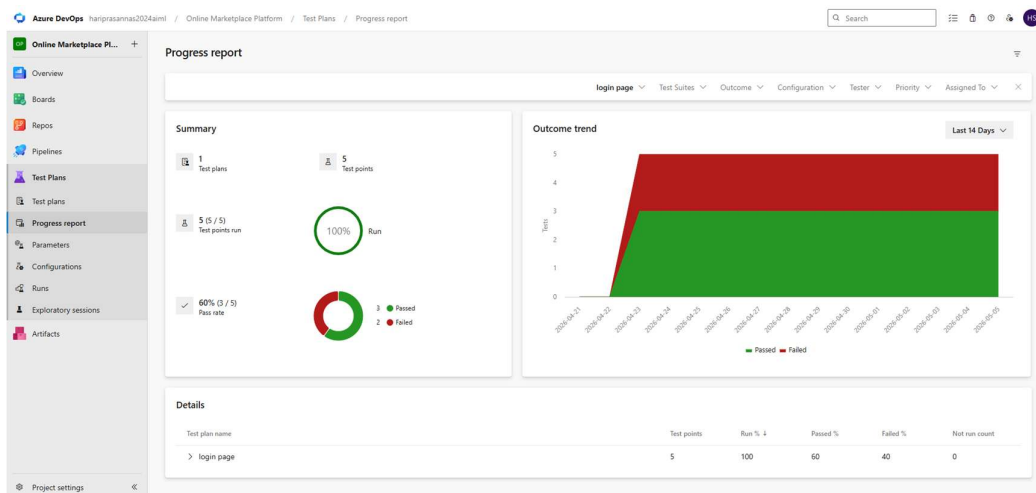
Test Results

8. Test report summary

9. Assigning bug to the developer and changing state

10. Progress report

11. Changing the test template



RESULT: The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path.